CodeBlooded: from customer data to the right next action

PTP-break prevention and next best action · Maple Bank synthetic data (snapshot 2026-09-28) · as built

SOURCES

Five source systems `data`

CRM · Cards · Lending · Core deposits · Collections (cases, promises, contacts, agent notes) · bureau file · 6 Parquet history tables · 25,000 call transcripts. **No shared customer key.**

Policy documents `data`

19 policy / procedure / FAQ documents with versions and authority levels.

Warehouse (DuckDB) `data`

`raw_*` typed, IDs as text → `cur_*` views (data traps fixed once) → `c360.duckdb`, `features.duckdb`, `decisions.duckdb`

L1 · Data product `rules`

Identity matching: CRM duplicates merged only on strong evidence; accounts linked by DOB + name + postal + phone, tiers A–E, per-field evidence, match confidence.

Shared-mobile block (phone never decisive alone) · **conflicts kept** side by side · **trust score** → golden **Customer 360** (1,000,729 customers, 93 fields) · data contract · quality report

Precision 99.84% · recall 97.4% (on collections-linked accounts)

L2 · Insight `rules` `LLM optional`

Refusal rules (protected attribute, proxy, PII, off-topic) → transcript summary → **certified SQL templates** → BM25 policy retrieval with citations → optional LLM planner (off) → refuse.

Every SQL passes a **sqlglot guard**: one SELECT, allow-listed views, no PII or protected columns, read-only, row cap.

Dev benchmark 21/21 · 35 answers, 6 refused

L3 · Features `rules`

One registry of SQL feature definitions, used for training and for live scoring (tested: batch = single-row).

Structured + text features from agent notes · point-in-time rules · each feature tagged *model / gate-only / excluded / diagnostic* · **audit blocks** protected, proxy and outcome columns.

L4 · Models & decisioning `1 ML model` `rules`

LightGBM, isotonic-calibrated: one number, the break risk (AUC 0.6445; below the incumbent's 0.712).

Ordered policy gates (protected block · hard stop · representative · case stage · dispute · **identity** · hardship · vulnerability · consent · contact budget · payment received) restrict what is allowed.

Rank actions (incl. **no action**) → review triggers → explanation: 3 reasons, evidence, what would change it, policy citations.

Decision Record `append-only`

Per recommendation: feature vector, gate inputs, model file hash, thresholds, holiday calendar, code commit, identity (C360). **Hash-chained**; verified on read.

Human review `person decides`

Accept · change · escalate, with a reason. Gates **re-checked at review time**: an override can never breach one. Hardship confirmation (reviewer roles only) writes a new, softer linked record. Every attempt, including refused ones, is logged.

Review triggers `risk-based`

Uncertain identity (C360 trust < 0.70) · possible hardship · low model confidence · policy conflict · high-impact action · insufficient data · fairness audit failure. About 14% of open promises need review.

API · `app/main.py` (FastAPI)

Thin layer: strict request schemas, structured errors, request ids. **Nothing in it decides anything.** Serves the built UI.

`/customers` · `/promises/upcoming` · `/recommend` · `/decisions/{id}/review` · `/audit/verify` · `/ask`

UI · React + Vite `agent desktop`

Agent queue (promises due soon, scored, riskiest first) · **Customer 360** (identity evidence, linked accounts, conflicts) · **Decision** (recommendation, why, gates, review panel, history) · **Ask the data** (answer + SQL or sources, or a refusal) · **Decisions & audit** (hash-chain check)

Governance, all layers: protected attributes excluded from the C360, the model and risk logic (audited) · hardship can only soften treatment · "no action" is a valid answer · the model supplies one number, everything that decides what may happen to a customer is rule-based and logged · PII and protected columns blocked in Insight · lineage raw → curated → C360 → feature → decision · reviewer roles are a demo directory, not real authentication.

☐ Data / services ☐ Deterministic rules ☐ AI / ML component ☐ Human ☐ Audit store ☐ Interface